In Dexterous Manipulation

STAM-FlashSAC

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Presented by Zheng-Lin Wu

What is STAM ?

What is STAM ?

Short **T**erm

Action-observation

Memory

Why **S**hort **T**erm **A**ction-observation **M**emory ?

A single frame is not enough

Current obs.

→ What it looks like now

A single frame is not enough

Current obs.
with obs. history

→ What it looks like now
What happened

A single frame is not enough

Current obs.
with obs. + action history

→ What it looks like now

What happened

What we did

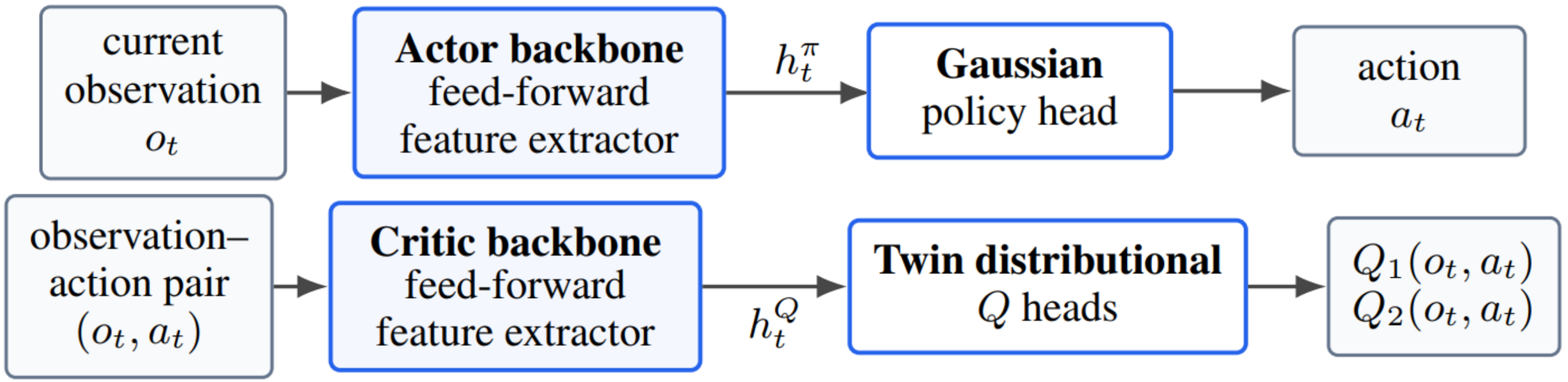
How we build this?

FlashSAC

Fast off-policy learning

Stable high-dimensional control

Strong feed-forward backbone



FlashSAC

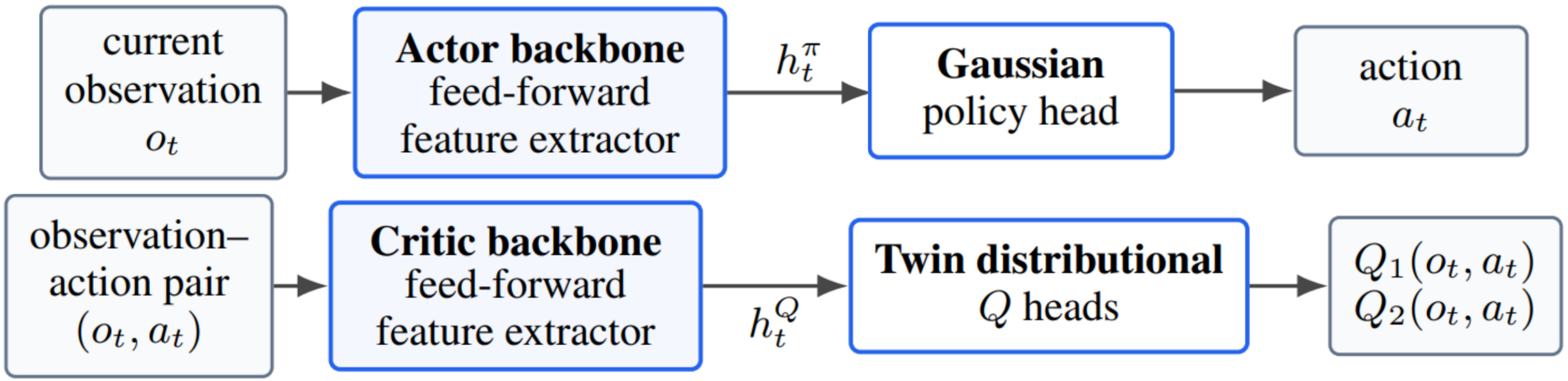
NTU Deep Reinforcement Learning Class Spring 2026 HW4 Q3 LeaderBoard

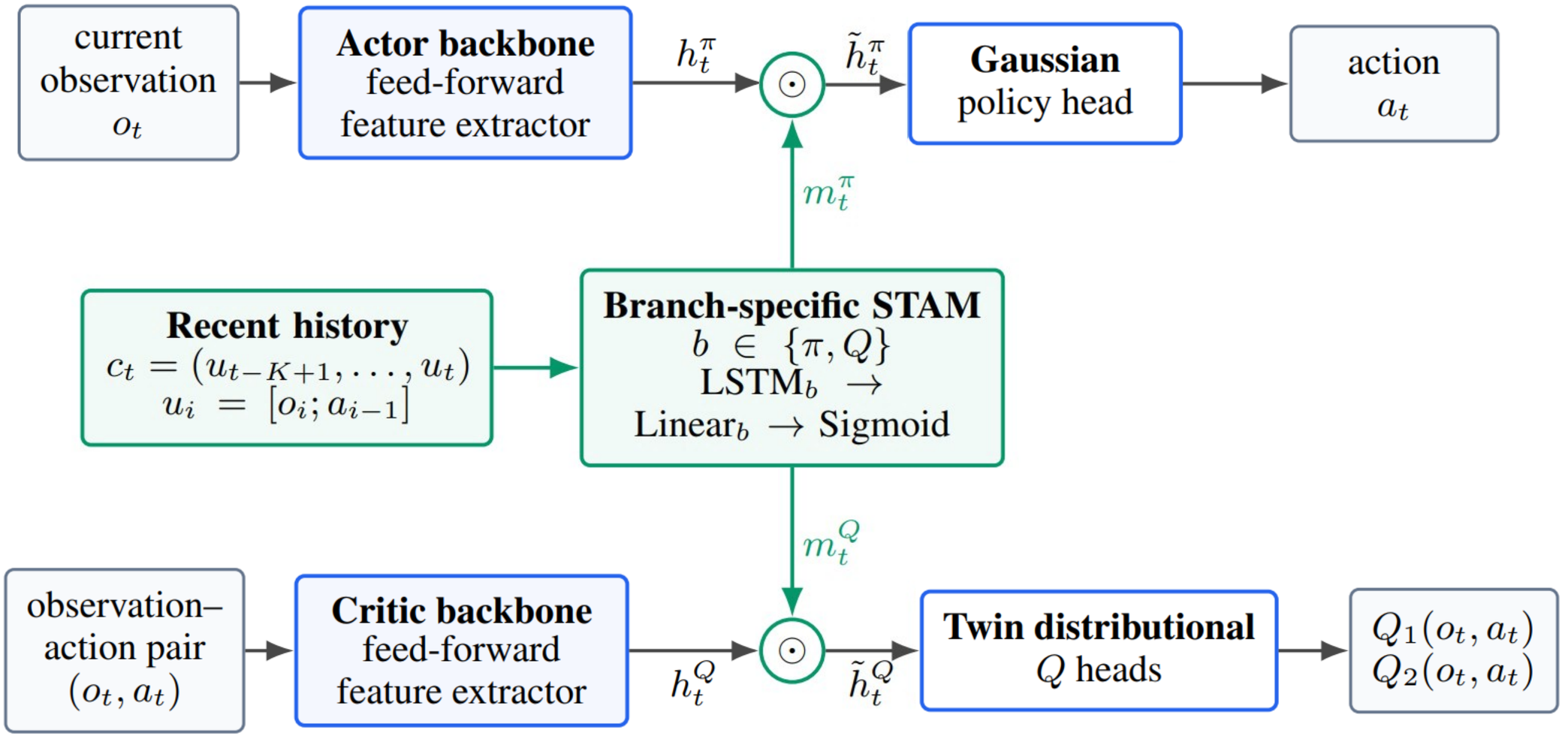
DMC · humanoid-run · 100 episodes

Score = mean(returns) − std(returns)

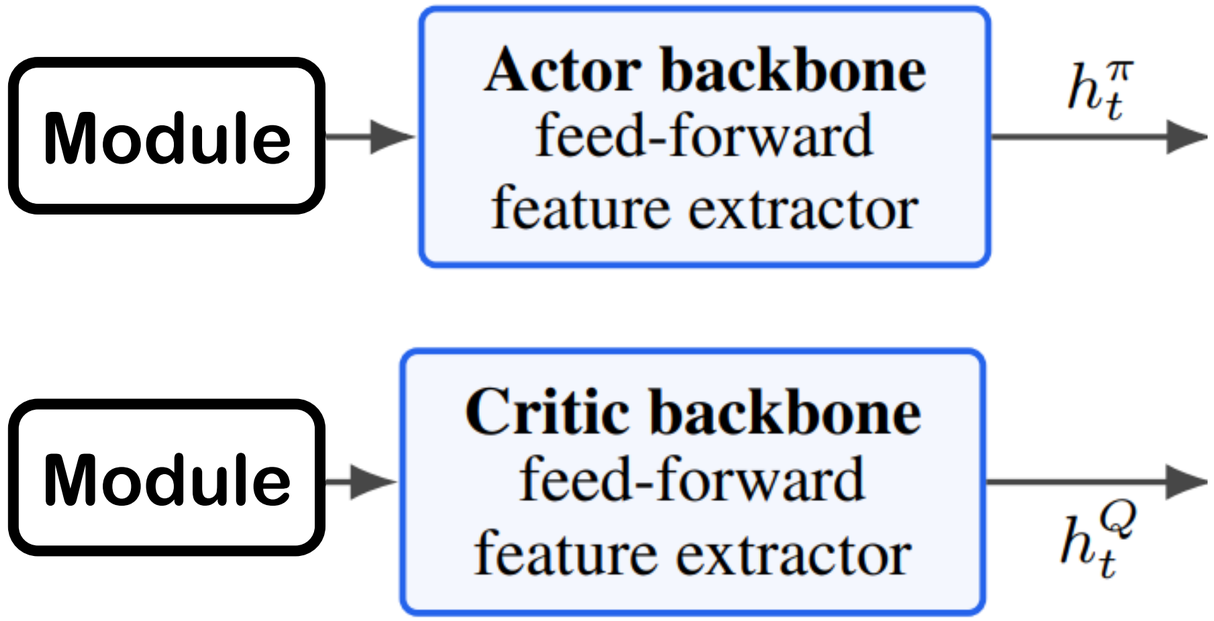
Last updated: May 6, 2026, 11:53 PM

#	STUDENT ID	SUBMITTED	SCORE	MEAN RETURN	STD RETURN	EPISODES
1	b12902071	5/3/2026 09:40 PM	846.78	881.03	34.25	100
2	b11902171	5/4/2026 01:12 PM	828.80	836.07	7.27	100
3	b11705045	5/3/2026 12:28 AM	803.71	825.71	22.01	100
4		4/29/2026 10:01 PM	701.73	731.28	29.55	100
5		4/27/2026 10:57 PM	643.10	690.96	47.85	100




current observation o_t
Actor backbone feed-forward feature extractor
h_t^π
$\tilde{h}_t^\pi$
Gaussian policy head
action a_t
m_t^π
Recent history
$c_t = (u_{t-K+1}, \ldots, u_t)$
$u_i = [o_i; a_{i-1}]$
Branch-specific STAM
$b \in \{\pi, Q\}$
$\text{LSTM}_b \rightarrow$
$\text{Linear}_b \rightarrow \text{Sigmoid}$
m_t^Q
observation–action pair (o_t, a_t)
Critic backbone feed-forward feature extractor
h_t^Q
$\tilde{h}_t^Q$
Twin distributional Q heads
$Q_1(o_t, a_t)$
$Q_2(o_t, a_t)$

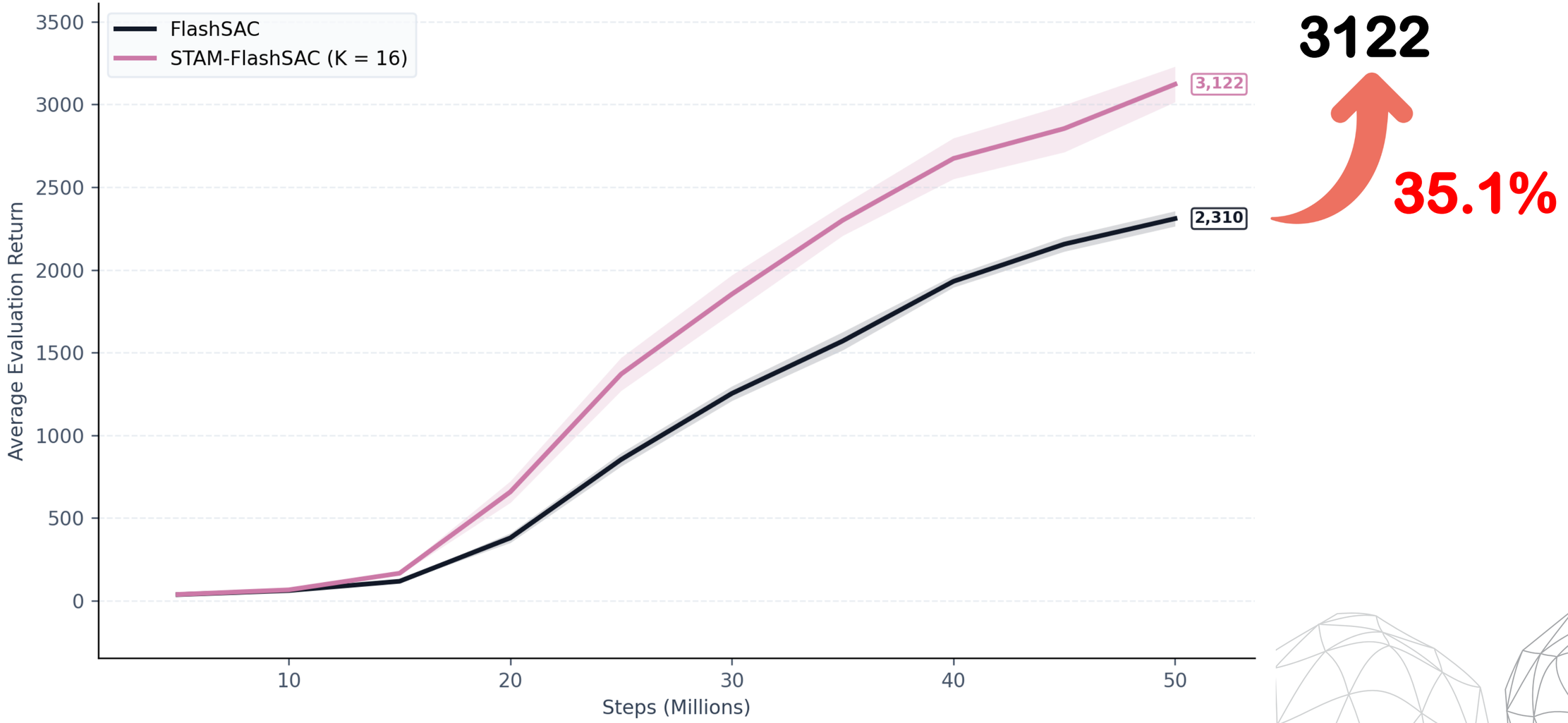
Why not a fully recurrent FlashSAC?



Do not disturb the FlashSAC backbone

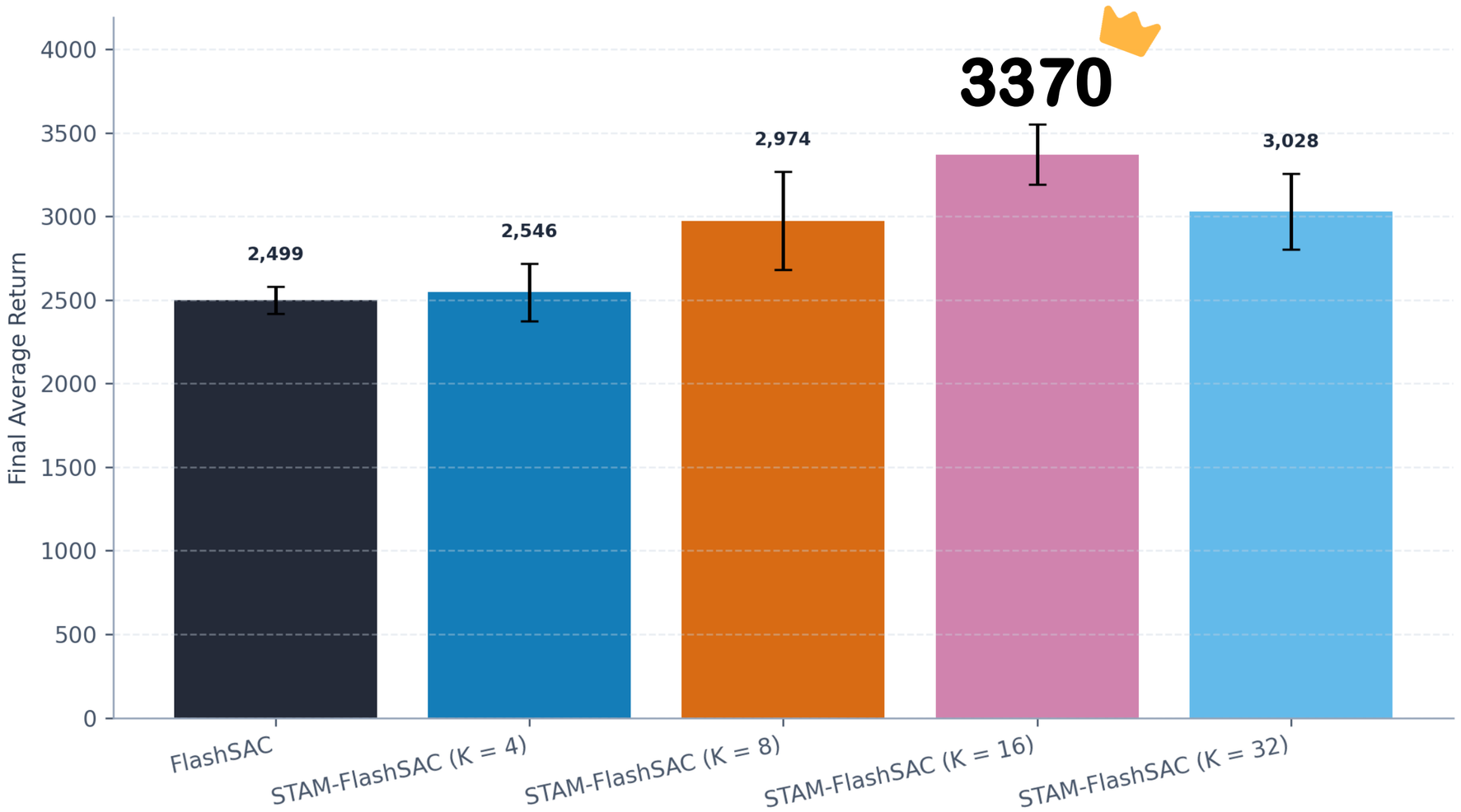
Does STAM actually help?

Env. : Shadow Hand OpenAI-FF # seed: 10



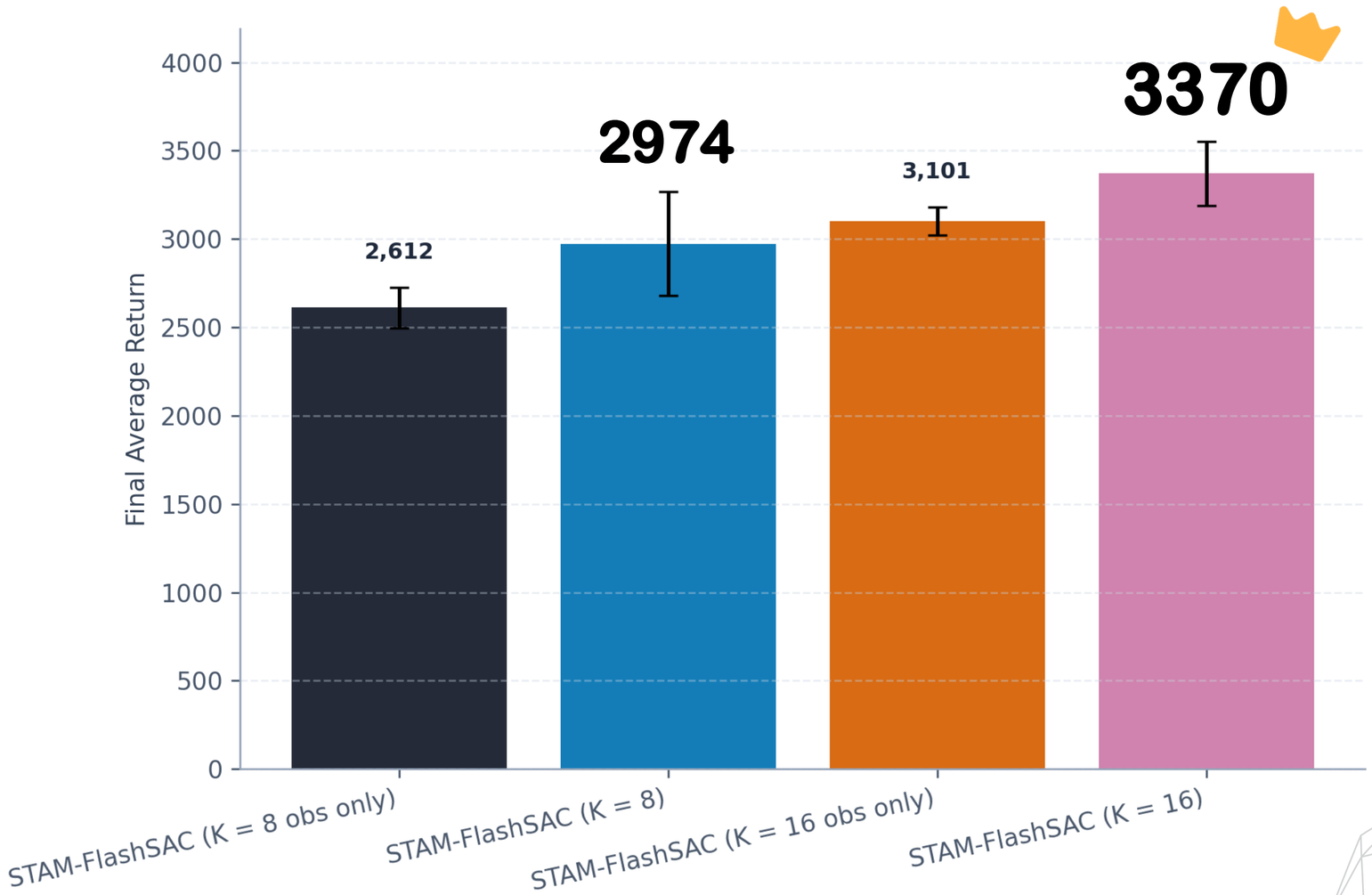
Is longer memory useful?

Env. : Shadow Hand OpenAI-FF # seed: 3



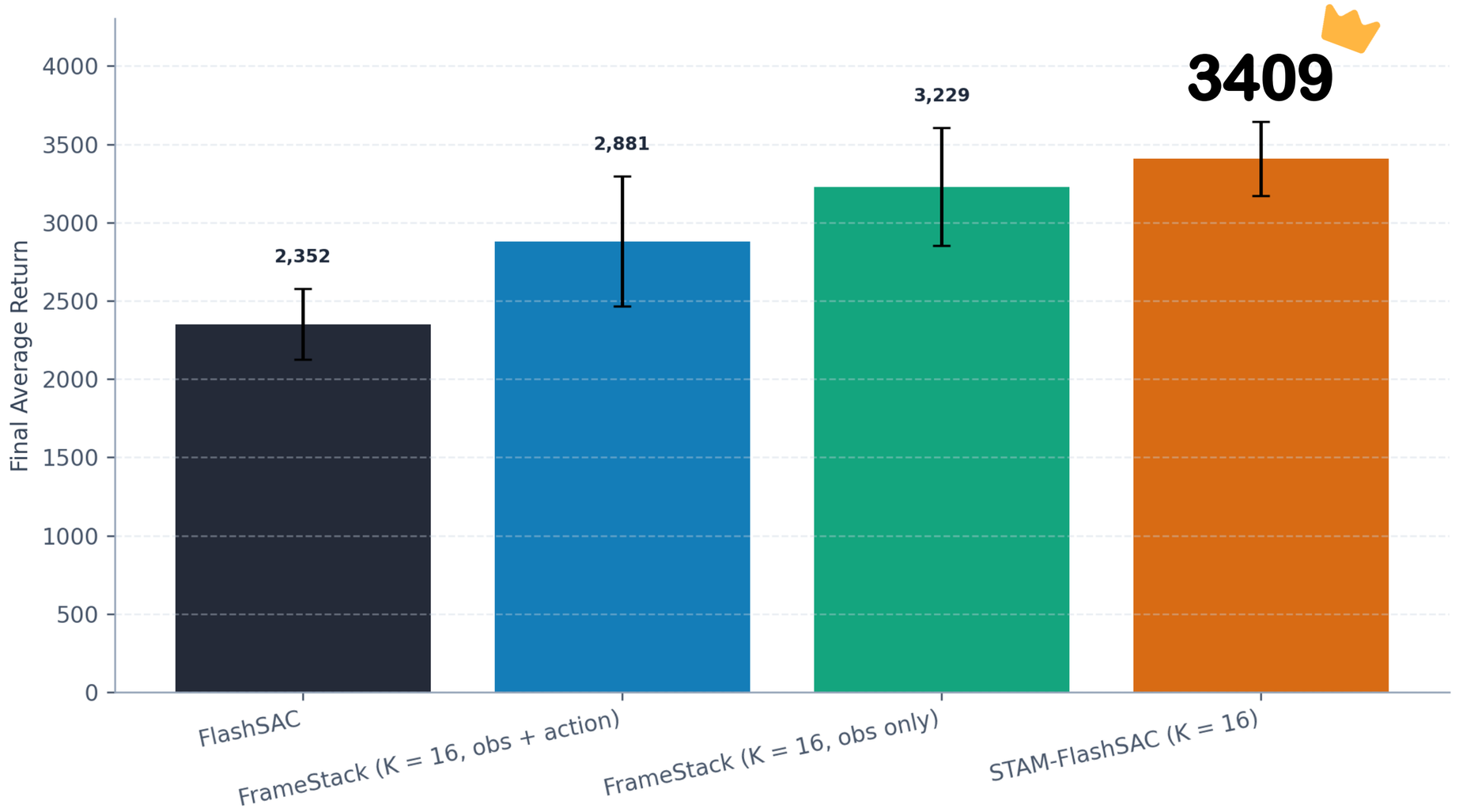
Do previous actions really matter?

Env. : Shadow Hand OpenAI-FF # seed: 3



Why not just stack frames?

Env. : Shadow Hand OpenAI-FF # seed: 5



Beyond the main result

Is STAM still lightweight?

→ wall-clock time + parameter count

Is the design choice important?

→ gating vs. concatenation

Where should memory be applied?

→ actor-only / critic-only / both

Does the encoder matter?

→ LSTM / GRU / MLP

Does it transfer to other settings?

→ Shadow Direct + Allegro Direct

Did we rule out confounds?

→ shuffled actions + n-step control

Future work

Broader task coverage

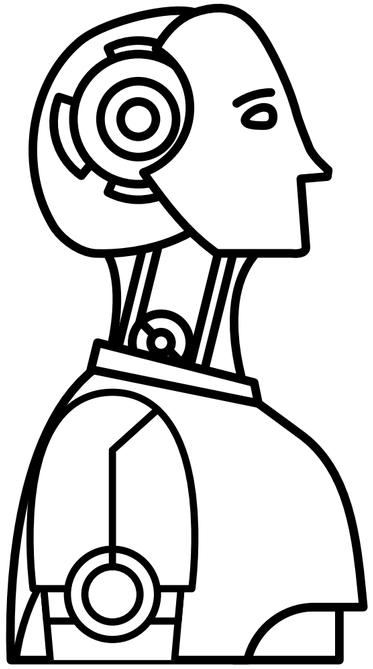
→ More hands, objects, and dexterous tasks

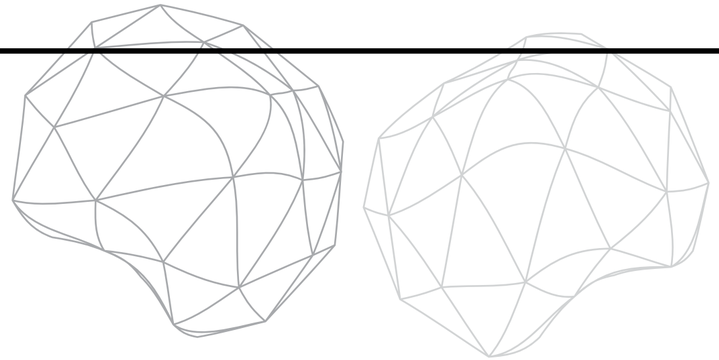
Better memory module

→ Validate GRU with more seeds and tasks

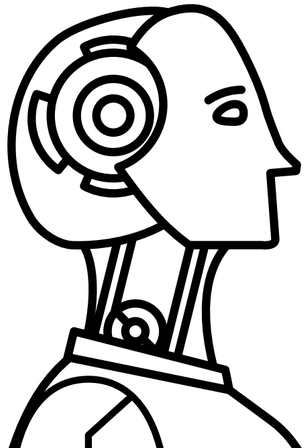
Adaptive memory

→ Learn how much history each state needs





Thank You For Your Attention



Reference

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